
ROBOCON '27'
P.O.W.E.R.
//DASH

CLARIFICATIONS

The most up-to-date version of the rules is available on the website. Vendors will be notified by email of any changes to the rules. The latest version of the rules will apply at the competition. Where the meaning or application of a rule is unclear, the RoboCon organisers will provide an official interpretation. Any official interpretation will apply equally to all Vendors. For questions or clarification, please email robotics@hrsfc.ac.uk.



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1. MISSON STATEMENT

It's the year 2028, and AI needs more power. Unfortunately, the world has run out of power stations - and in the spirit of reusing resources (and making as much money as possible) the global coalition of power companies has unanimously decided to restore nuclear disaster sites as modern power stations. 5 sites have been selected - Chernobyl, Fukushima, Windscale, Three-Mile Island, and the lesser-known Bugey, which had a tragic meltdown in the summer of 2027. This decision was made after a long and hefty debate (consulting Claude).

Since technically no one owns these old power stations, any power company can start fuelling them in exchange for shares in the profits. However, the major issue concerning these power stations is that they are within nuclear fallout zones. For obvious reasons, this makes using them a bit tricky.

After another long and hefty debate, another proposal was made - 'Robotic Orb Bearing Offline Transports' (ROBOTs) could be used to deliver fuel to the power stations. While ROBOTs are not entirely immune to ionising radiation, it is more ethical and cheaper to sacrifice ROBOTs than humans.

Power companies are hiring independent contractors to design ROBOTs for this task. The ROBOTs must deliver 'energy orbs' to the disaster sites. The greater the amount of 'Nuclear Refuelling Globe' NRG the contractor is able to supply to the power stations, the greater the share in the stations' profits.

The UN is organising a set of field trials to evaluate the suitability of ROBOTs by organising a series of 'Profitable Orb-delivery With Expendable Robotics Delivered Autonomously Sans Humans' (POWERDASH) events.

Four different major power companies have taken the initiative to compete for these valuable shares – Flemish Gas (Red), Mollusc (Yellow), Kraken Fission (Green) and Cheval (Blue). You are competing with other contractors to be commissioned by these companies to deliver these orbs. Your ROBOTs will start by competing in proof-of-concept competitions to demonstrate their capabilities for delivering orbs. The winners of these competitions will be the ones who will be commissioned by the power companies.

Trials will begin April 7th 2027.

2. AWARDS

2.1. **Alpha, Beta, Gamma Awards**

Awarded to the Vendors finishing in 1st, 2nd and 3rd place in the competition, respectively.

2.2. **Outside the Zone Award**

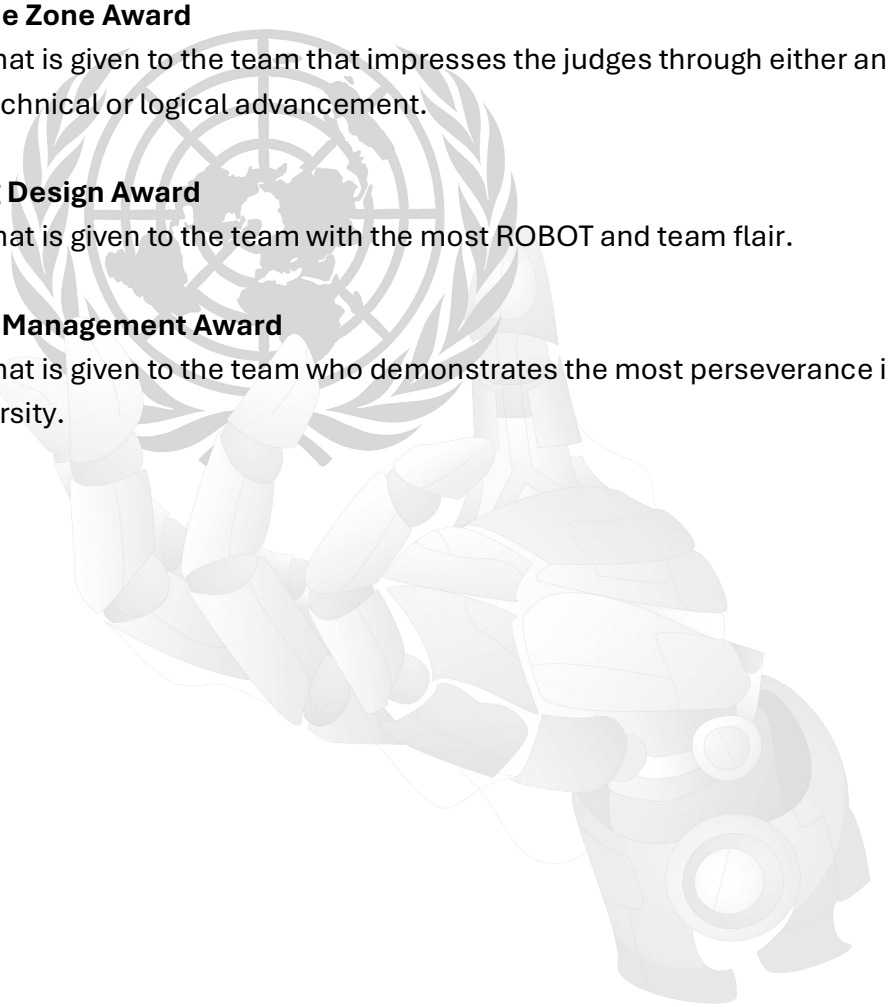
The award that is given to the team that impresses the judges through either an ingenious technical or logical advancement.

2.3. **Irradiating Design Award**

The award that is given to the team with the most ROBOT and team flair.

2.4. **Meltdown Management Award**

The award that is given to the team who demonstrates the most perseverance in the face of adversity.



3. SPECIFICATION

3.1. ARENA DIAGRAM

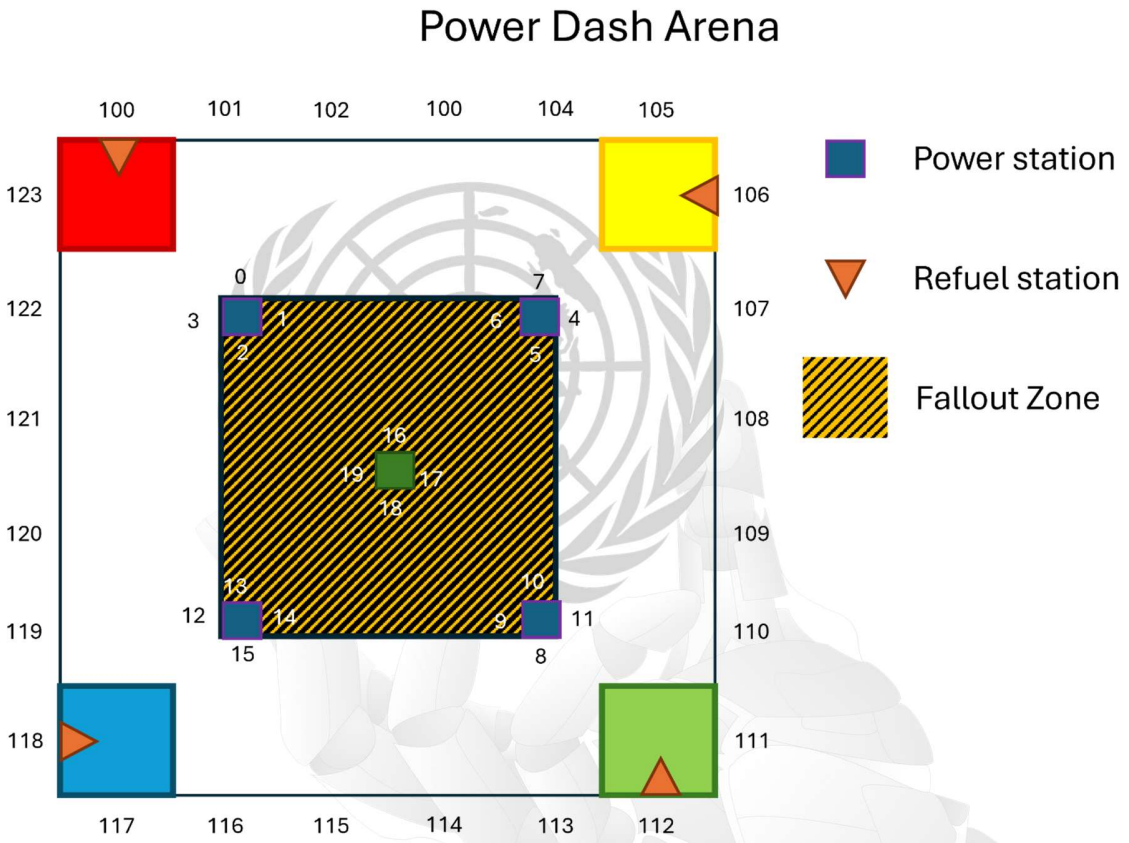


Figure 1

3.2. GAME ELEMENTS

3.2.1. Arena

3.2.1.1. The arena measures 6000 mm × 6000 mm (±100 mm).

3.2.1.2. The arena floor is covered with 500 mm × 500 mm close-pile industrial carpet tiles.

3.2.1.3. Each corner contains a 1000 mm × 1000 mm ‘Start Zone’ (±20 mm).

3.2.1.4. The central 3000 mm × 3000 mm (±20 mm) of the arena is the ‘Fallout Zone’.

3.2.1.5. The boundaries of each ‘Start Zone’ and the ‘Fallout Zone’ are marked with 48 mm-wide tape.

3.2.1.6. Each wall has six 200 mm × 200 mm markers, as shown in Figure 1 (page 7). The leftmost marker on each wall also serves as the marker for the corresponding refuel station.

3.2.1.7. The arena wall is approximately 500mm in height.

3.2.2. NRG Orbs

3.2.2.1. NRG Orbs are spherical objects with a diameter of 40 mm (±1 mm).

3.2.2.2. Each corner uses a different colour of NRG Orb.

3.2.2.3. Each ROBOT starts with two NRG Orbs onboard.

3.2.2.4. Additional NRG Orbs can be collected by visiting Refuel Stations.

3.2.3. Refuel Stations

3.2.3.1. Each corner has one Refuel Station, identified by the leftmost marker on its wall, as shown in Figure 1.

3.2.3.2. Each Refuel Station dispenses only the NRG Orb colour assigned to its corner.

3.2.3.3. The refuel station generates NRG Orbs at a rate of 1 every 20 seconds.

3.2.3.4. Each Refuel Station has a 400 mm × 400 mm button mounted on the wall, with its front face 10 mm (±5 mm) from the wall.

3.2.3.5. A NRG Orb is released within 2 seconds of the button being pressed.

3.2.3.6. Each press releases no more than 1 NRG Orb.

3.2.3.7. Each Refuel Station releases NRG Orbs from a height of 450 mm (±20 mm) above the floor and at a horizontal distance of 200 mm (±20 mm) from the wall centred on the button.

3.2.4. Power Stations

3.2.4.1. The Power Station in the centre of the ‘Fallout Zone’, Chernobyl, can handle only 1 NRG Orb per trial.

3.2.4.2. Power Stations in the corners of the ‘Fallout Zone’, Three Mile Island, Fukushima, Windscale, Bugey, can handle 4 NRG Orbs per trial.

3.2.4.3. When a Power Station reaches its maximum capacity, any additional NRG Orbs are ejected into the ‘Fallout Zone’.

3.2.4.4. Each Power Station face is marked with an 80 mm × 80 mm marker.

3.2.4.5. Power Stations are refuelled by placing NRG Orbs on top of the Power Station.

3.2.4.6. Power Stations measure 270 mm × 270 mm × 150 mm (±20 mm), approximately the width and height of a 12-litre Really Useful Box.

3.3. MARKERS

3.3.1. Each marker encodes a unique number that ROBOTs may use to navigate the arena.

3.3.2. The markers are printed in black and white and laminated with a matt finish. They can be downloaded from the documentation section of the RoboCon website at www.robocon.uk.

3.3.3. The arena walls are marked with AprilTags numbered 100–123, as shown in Figure 1.

3.3.4. Power Stations are marked with AprilTags numbered 0–20, as shown in Figure 1.

3.3.5. Each Refuel Station is identified by the AprilTag assigned to its corner, as listed below.

Corner	**AprilTag ID**
Flemish Gas (0)	100
Mollusc (1)	106
Kraken Fission (2)	112
Cheval (3)	118

4. REGULATIONS

4.1. GENERAL

4.1.1. The trials will be adjudicated by the Director General.

4.1.2. Hills Road RoboCon reserves the right to examine any ROBOT's software and hardware at any time, to verify that rules are being followed.

4.1.3. Assistance from Hills Road RoboCon is provided with no guarantees.

4.1.4. Attaching a GoPro or similar small video recorder to your ROBOT to record a trial is permitted, but it must be powered by its own internal batteries and cannot be connected to any of the other electronics of the ROBOT. It must fit within the internal dimensions defined in Paragraph 4.3.3.

4.1.5. A USB port must remain free for the use during trials.

4.1.6. The Brainbox is supplied as a sealed unit. It must not be tampered with. If you need access to the contents of the Brainbox, you must contact us so we that can help you with your plans.

4.2. TRIAL CONSTRAINTS

4.2.1. ROBOTs may not deliberately damage anything. This includes the arena, supply crates, and other ROBOTs. At the Director General's discretion, Vendors who engage in collisions deliberately or do not take sufficient precautions to avoid collisions may be disqualified from trials until the issue has been resolved. The Director General decides whether or not this action was deliberate.

4.2.2. Vendors must not imitate any features of the arena in a way that could potentially confuse other Vendors' ROBOTs.

4.2.3. No remote-controlled systems are permitted. ROBOTs may not include additional radio transmitters or receivers to those in the BrainBox.

4.2.4. ROBOTs may not deliberately leave debris in the arena.

4.2.5. The ROBOT is to be started in the arena using a physical start button. ROBOTs must be placed in the arena so that one of the vendor's representatives can reach it from outside of the arena.

4.2.6. While accidental contact between ROBOTs may occur, ROBOTs should not intentionally interact with other ROBOTs - we reserve the right to review your code to ensure it complies with this rule.

4.2.7. ROBOTs must not attempt to retrieve power orbs once they have been delivered to a Power Station. This is an inherently unsafe and dangerous procedure.

4.3. SAFETY AND INSPECTION

4.3.1. ROBOTs must pass a safety inspection by a RoboCon safety inspector before they are permitted in the arena.

4.3.2. The lithium polymer battery provided in the kit must be shielded from mechanical and thermal harm. This includes ensuring that it is protected from harm in the case of accidental collision with another ROBOT. Vendors found to be in violation of this rule will have their batteries confiscated until they have demonstrably rectified the issue.

4.3.3. At the beginning of each trial, ROBOTs must fit into a cube with internal dimensions of 400 x 400 x 400 mm. During the trial, the ROBOT may extend beyond this size.

4.3.4. The ROBOT must have a RED power switch which must be easily accessible while the ROBOT is active.

4.3.5. You may use custom hardware to enhance your ROBOT's electronics, but all power should be drawn from the connectors on the BrainBox.

4.3.6. All wires connected to the ROBOT's ground (0V line) must be black. Black wires must not be used for anything else. It is strongly recommended that all wiring is neat and removable, as this will reduce the time required to debug problems, and Vendors may be asked to tidy their wiring before a member of Hills Road RoboCon will approach any issues with their ROBOT.

4.3.7. All electronics should be securely fixed to the ROBOT (not able to be shaken loose) and should also be easily removable if required.

4.3.8. It must not be possible to injure oneself on the ROBOT. This may be tested using a Frankfurter sausage to simulate a finger – anything that could cause harm, such as high-speed rotating parts, must be suitably shielded.

4.3.9. All power to ROBOT mechanisms must come from the 12V battery supplied.

4.3.10. If Vendors wish to use batteries, chargers, or cables other than the ones provided with the kit, they must seek approval from Hills Road RoboCon through robotics@hrsfc.ac.uk first.

5. TRIALS

5.1. TRIAL ORGANISATION

5.1.1. Contracting Vendors will be given a published schedule for when they are to be representing each power company.

5.1.2. Vendors tendering to each of the four power companies will be invited to each trial.

5.1.3. Vendors will be given no recognition for missed trials.

5.1.4. Each trial lasts for 180 seconds and after which ROBOT will deactivate.

5.1.5. A trial may be ended prematurely if all participating contractors state to the Director General that the trial has been completed to the best of the ROBOT ability.

5.1.6. Humans must evacuate to behind the protective wall before the trial commences.

5.1.7. Humans must not in any circumstances enter any body part or other objects past the protective wall during a trial.

5.1.8. Humans must not in any circumstances interfere with the protective wall, the functioning of the refuel station, or ROBOTs that are competing in the trial. Contractors who interfere with the trials will be asked to be leave and may be disqualified.

5.2. TRIAL EVALUATION

5.2.1. Evaluation of contracting Vendors will begin 10 seconds after the end of a trial.

5.2.2. At the end of the trial, the Director General will total the score for each competing vendor.

5.2.3. Scoring is judged according to the state of the arena at the end of the trial.

5.2.4. Vendor representatives must not enter the arena or touch any ROBOTs or game items in the arena while the vendor score is being calculated in order to ensure fair scoring. Touching any ROBOT or item before the Director General 'releases' the arena may result in the offending team's league points for the trial being forfeit.

5.2.5. The following points are available once per trial

Action	Points
ROBOT leaves its Start Zone	+1
ROBOT collects an NRG orb from a refuel station	+2

5.2.6. At the end of the trial, NRG orbs from each power company will be allotted points as following

Location	Points	Total Capacity
Chernobyl	+6	1
Three Mile Island Fukushima Windscale Bugey	+3	4
Fallout zone	+1	-

5.2.7. In the event that NRG orbs from one power company are placed in multiple power stations, the score from 5.2.5. will be multiplied by the number of occupied stations.

Fuelled Stations	Points Multiplier
0	1x
1	1x
2	2x
3	3x
4	4x
5	5x

6. LEAGUE POINTS AND KNOCKOUT TRIALS

6.1. LEAGUE POINTS

6.1.1. Only Vendors with ROBOTs present in the area at the start of a trial can score league points from that trial.

6.1.2. League point scoring:

Score	League points
Highest score	8 League points
Second highest score	6 League points
Third highest score	4 League points
Fourth highest score	2 League points

6.1.3. In the case of a draw, in which two or more Vendors score the same, each of the Vendors will gain the average number of points of their places. For instance, if two Vendors score equally and are therefore in joint first place, they will both score the mean of the first and second place scores – 7 league points.

6.1.4. Vendors whose ROBOTs were disqualified from a trial will earn no league points for that trial.

6.2. KNOCKOUT TRIALS

6.2.1. Once the league has been completed, a knockout competition will begin. The positions of the Vendors in the league will seed the positions of Vendors in the knockout trials.

6.2.2. Each trial in the knockout competition involves up to 4 Vendors. The Vendors that come 1st and 2nd in each knockout trial will continue to the next trial of the knockout.

6.2.3. In case of a tie in a knockout trial, the team ranked highest in the league will go through.

6.2.4. If there is a tie in the final for first place, then a rematch may be played at the Director General's discretion.

7. KIT RETURN

Each team will be provided with a kit which contains a disclaimer form detailing your obligations with respect to assembly, use and return of the kit, which is lent to you for the duration of RoboCon 2027. Each team is issued with a kit with the following parts, which is to be returned at the end of the competition.

- Items to be returned:
  - Electronic kit:
    - 1x Brainbox
    - 1x Power switch – latching (red switch, black connector)
    - 1x Start button – momentary (black switch, green connector)
    - 1x 6-pin GPIO connector
    - 1x 2-pin 12V Accessory connector
  - Batteries:
    - 1x 3S (11.1V nominal) LiPo Battery
    - 1x Turnigy E3 Compact 2S/3S LiPo charger
    - 1x "LiPo Safe" bag for storage and charging of batteries
  - Computer:
    - Linux laptop, paired with electronics for programming and Wi-Fi download
    - Power Brick and lead
  - Other:
    - 1x 12L Really Useful Box and Lid
- Items to be returned if in working order:
    - 1x Minibot chassis
    - 2x TT motors with connectors
    - 1x Microservo 9g SG90
    - 1x Inert NRG Orb

If possible, the kit should be returned at the competition, but in no case should it be returned later than 14 days after the competition. If you wish to keep the kit beyond that, this must be arranged prior to the competition date via an email to robotics@hrsfc.ac.uk

THANKS

We would like to thank Will Munns, Mr Smalley, Mr Massey and the sponsors listed below, without whom we would not have been able to run this competition.



"Robotics, cybernetics and artificial intelligence are some of the most rapidly changing fields of science and technology, with tremendous opportunities for future engineers. This generation of students are the ones who will write the next chapter in technology, and this competition may be the starting point for their careers. Several Hills Road students who became interested in engineering and robotics through the Robotics group and entering competitions have gone on to degrees and careers in this exciting field",

- David Massey, Founder of Hills Road Robotics



Hills Road
Sixth Form College
Cambridge



EXECUTIVE ORDER

Our top genius scientists they're coming to me with tears in their eyes saying, "Mr president sir, Chernobyl has gone critical". They're calling it a disaster, a disaster like you've never seen before it's too much energy, can you believe that? First they say not enough, I fix it, and now they say too much! They're tired of winning, are you tired of winning? I'm not, that's why when they called me because they don't know what to do I told them, because no one knows nuclear like I do, that we need to use more of my Big Beautiful Balls. Blow the radiation back onto the ground with a big beautiful ball delivered right in the middle. And they say to me "Mr president you're a genius sir" I know, you know and now they know.

The deepstate at the EPA would tell you "turn it off" and "think of the environment" but the look at the economy we've never numbers like this before.

I'm signing this executive order to give you more points than you've ever seen, 9 points, they can't even count that high, 9 points, the most points ever. If you can deliver a big beautiful ball to the funnel cloud.

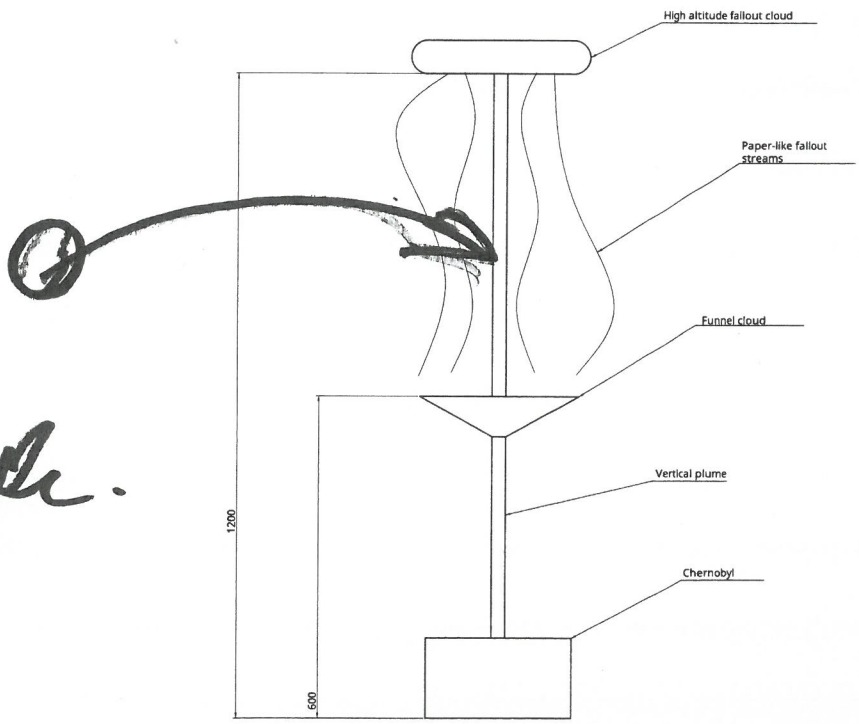


Figure 1: Unexplained phenomenon at chernobyl site following preliminary testing of POWERDASH

The White House